

Cyanobacteria

PRECAMBRIAN: 4,000–542 MYA

Cyanobacteria are among the oldest and simplest organisms alive today. Sediment has built up in round clusters around a cyanobacteria nucleus.

**Stromatolites**

PRECAMBRIAN: 4,000–542 MYA

Stromatolites are layers of sediment that were deposited by microorganisms in the coastal shallows of ancient seas. Living stromatolites still exist today.

**Calymene trilobite**

SILURIAN: 444–416 MYA

Early scholars called trilobite fossils “petrified insects.” They were identified as arthropods in the eighteenth century.

Clavilithes shell

PLIOCENE: 5.3–1.8 MYA

Clavilithes is related to the modern whelk. Shell fossils may be found far from the sea, leading Greeks to deduce that land was once covered by water.

**Harpoceras ammonite**

LOWER JURASSIC: 199.6–175.6 MYA

The Roman naturalist Pliny named the ammonite fossil after the ram horns of the Egyptian god Ammon.

**Pentacrinites crinoid**

PALEOGENE: 65–23 MYA

Delicate animals such as crinoids could be beautifully preserved when the soft mud in which they lived turned into rock.

FOSSILS

SINCE THE EARLIEST TIMES, FOSSILS HAVE HELPED US UNDERSTAND THE HISTORY OF LIFE ON EARTH

Fossils have been known to man since prehistory, but early Christian scholars believed they were the remains of creatures that died in the Biblical flood. We now know they are evidence of evolution on a changing planet.

The scientific study of fossils has its roots in the Age of the Enlightenment, when naturalists began to describe and classify their finds. By the 1800s, geologists recognized not only that rock layers built up over time, but also that each layer had a distinctive set of fossil types. Supporting evidence was recovered from this fossil record to support Charles Darwin's theory of evolution. Today, paleontology routinely uses technology that can determine the age of fossils and even study their DNA.

Temnocidaris sea urchin

CRETACEOUS: 145–65 MYA

Animals with hard parts such as spines or shells can leave abundant fossils. By 1850, geologists could assign rock layers to periods of time, according to the fossils they contained.

**Halysites tabulate coral**

SILURIAN: 444–416 MYA

Honeycomb-like tabulate corals were fossilized in Silurian rock layers. Tabulate corals existed in colonies and were an important component of Silurian shallow-water fauna.

**Hydrophilus water beetle**

PALEOGENE: 65–23 MYA

Wet mud is good for preserving animals. The first fossilized beetle specimens—hard wing cases—were found in the 19th century. Later, entire insects were found too.

**Dolomedes spider**

PALEOGENE: 65–23 MYA

Today, *Dolomedes* spiders inhabit wetlands, but fossils in amber show they once lived on trees and were trapped in drops of resin that hardened over time.

